Complete genome sequences of three co-occurring *Pseudomonas* isolates from California Botanic Garden topsoil

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Abstract:

We report the complete genome sequences of three Pseudomonas isolates recovered from topsoil at the California Botanic Garden. Two isolates share ~99.4% ANI, enabling investigation of intraspecies microvariation; the third represents a co-occurring distinct species, together capturing species- and strain-level genomic diversity within a natural soil community.

Announcement:

High-fidelity, complete genomes of closely related environmental isolates, such as those of the abundant and diverse *Pseudomonas* genus[1], provide a valuable resource for studying micro diversity at full genome resolution. Here, we report the complete genome sequences of three Pseudomonas isolates recovered from soil samples at the California Botanic Garden (Claremont, CA).

Soil samples were collected using a T-Handle Soil Sampler Probe Rod Tool (Lichtund); see table 1 for details of collection sites. Soil was stored in Ziploc bags and transported to Keck Graduate Institute (KGI) and stored at 4˚C until further processing. Bacterial enrichments were performed using 5 g of soil in 25 mL of Pseudomonas Minimal Media (2 mM $MgSO_4$, 0.31 mM $FeCl_3$, 0.03 mM $CaCl_2$, 15 mM NaH_2PO_4, 15 mM K_2HPO_4, pH adjusted to 6.8) amended with 5 drops of both Na-Succinate (10% solution) and NH_4Cl (20% solution) as carbon and nitrogen sources [2]. After enrichment for two days, we plated the enrichments using a 1 uL inoculation loop on Pseudomonas Minimal Agar (1.5% agar). After two days of growth, colonies were picked and plated on Pseudomonas Isolation Agar (PIA, Criterion No:C6681; 500 mL milliQ water, 12.5 g PIA base medium, 10 mL glycerol). Single colonies were picked and grown overnight in lysogeny broth (LB; 25 g/L, 2.5% w/v) medium. Next morning, cultures with an optical density (OD600) > 0.6 were selected for storage. 650 uL of selected culture were taken and mixed with 50% glycerol solution in a 1:1 ratio, and these prepared samples were stored as stocks at -80˚C.

For genome sequencing, colonies were grown from stocks on LB agar (1% agar) and then grown in 10 mL of LB media (2.5% LB). Overnight cultures were then pelleted at 3000 g for 15 minutes and washed twice with PBS. DNA was extracted using the QuickDNA Kit (Zymo Research Corporation), following manufacturer's instructions. DNA was quantified using Portlite Fluorimetric DNA Quantification Kit (AAT Bioquest) and renormalized to 50 ng/uL. 20 uL were sent

out for sequencing at Plasmidsaurus (Eugene, OR) for Nanopore Sequencing using the ONT ligation sequencing kit v.14 (SQK0LSK1114) library preparation without size selection.

Default parameters were used for all software described in this study unless otherwise specified. Basecalling was performed with Dorado using the super-accuracy (sup) basecalling model (v4.3.0 [3]). Reads were quality-filtered with Filtlong (v0.2.1 [4]) using the flags "--min_length 300 --keep_percent 95". A de novo assembly was generated with Autocycler (v0.2.1[5]) from input assemblies produced by Flye (v2.9.6 [6]), hifiasm ([7]ONT mode), and Plassembler (v1.8.0 [8]). The circular assembly graph was reoriented using dnaapler (v1.2.0[9]). Genome completeness and contamination were assessed with CheckM (v1.2.2[10]). Isolates were taxonomically classified with sourmash (v4.9.4 [11]) against a GTDB database, which assigned all isolates to the genus *Pseudomonas*, two with an average nucleotide identity (ANI) of 99.5–99.75% to *Pseudomonas* sp013179515 (GCF_013179515.1) and one to *Pseudomonas capeferrum* (GCF_000731675.1, ANI 98.66 %). Genomes were annotated with Bakta (v1.12.0 [12]). Raw sequencing data and general genomic features are summarized in Table 1.

Data availability:

The complete genome of our isolates and raw sequencing data is available at PRJNA1482863 and PRJNA1483221 respectively. Individual accession numbers are listed in Table 1.

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References

[1] X. Wu, S. Monchy, S. Taghavi, W. Zhu, J. Ramos, and D. van der Lelie, "Comparative genomics and functional analysis of niche-specific adaptation in *Pseudomonas putida*," *FEMS Microbiol. Rev.*, vol. 35, no. 2, pp. 299–323, Mar. 2011, doi: 10.1111/j.1574-6976.2010.00249.x.

[2] K. Sandman and C. Ecker, "Pseudomonas Isolation and Identification: An Introduction to the Challenges of Polyphasic Taxonomy," *J. Microbiol. Biol. Educ.*, vol. 15, no. 2, pp. 287–291, Dec. 2014, doi: 10.1128/jmbe.v15i2.754.

[3] "GitHub - nanoporetech/dorado: Oxford Nanopore's Basecaller.," https://github.com/nanoporetech/dorado/.

[4] Wick R.R., " Filtlong. github.com/rrwick/Filtlong.," 2018.

[5] R. R. Wick, B. P. Howden, and T. P. Stinear, "Autocycler: long-read consensus assembly for bacterial genomes," *Bioinformatics*, vol. 41, no. 9, Sep. 2025, doi: 10.1093/bioinformatics/btaf474.

[6] M. Kolmogorov, J. Yuan, Y. Lin, and P. A. Pevzner, "Assembly of long, error-prone reads using repeat graphs," *Nat. Biotechnol.*, vol. 37, no. 5, pp. 540–546, May 2019, doi: 10.1038/s41587-019-0072-8.

[7] H. Cheng *et al.*, "Efficient near-telomere-to-telomere assembly of nanopore simplex reads," *Nature*, Feb. 2026, doi: 10.1038/s41586-026-10105-6.

[8] G. Bouras, A. E. Sheppard, V. Mallawaarachchi, and S. Vreugde, “Plassembler: an automated bacterial plasmid assembly tool,” *Bioinformatics*, vol. 39, no. 7, Jul. 2023, doi: 10.1093/bioinformatics/btad409.

[9] G. Bouras, S. R. Grigson, B. Papudeshi, V. Mallawaarachchi, and M. J. Roach, “Dnaapler: A tool to reorient circular microbial genomes,” *J. Open Source Softw.*, vol. 9, no. 93, p. 5968, Jan. 2024, doi: 10.21105/joss.05968.

[10] D. H. Parks, M. Imelfort, C. T. Skennerton, P. Hugenholtz, and G. W. Tyson, “CheckM: assessing the quality of microbial genomes recovered from isolates, single cells, and metagenomes,” *Genome Res.*, vol. 25, no. 7, pp. 1043–1055, Jul. 2015, doi: 10.1101/gr.186072.114.

[11] L. Irber *et al.*, “sourmash v4: A multitool to quickly search, compare, and analyze genomic and metagenomic data sets,” *J. Open Source Softw.*, vol. 9, no. 98, p. 6830, Jun. 2024, doi: 10.21105/joss.06830.

[12] O. Schwengers, L. Jelonek, M. A. Dieckmann, S. Beyvers, J. Blom, and A. Goesmann, “Bakta: rapid and standardized annotation of bacterial genomes via alignment-free sequence identification,” *Microb. Genom.*, vol. 7, no. 11, Nov. 2021, doi: 10.1099/mgen.0.000685.

Table 1. Genome assembly, annotation, and collection metadata for the three Pseudomonas isolates. Isolates are columns; metrics are rows. Coordinates in degrees-minutes-seconds (WGS84); elevation in meters. Completeness/contamination by CheckM; closest species and ANI by sourmash (GTDB).

	Isolate 8-5-2	Isolate 9-2-1	Isolate 8-10-2
Isolate (internal name)	8-5-2	9-2-1	8-10-2
Sequencing sample ID	B9TF7H_1_Sample_1_8A	XH254M_1_sample_1	CZJZZ4_1_rerun_Sample_2_8B_2
Represents (≈100%-ANI clones)	8-5-2	9-2-1	8-10-2
Genome accession (GenBank)	SAMN61139397	SAMN61139396	SAMN61139398
Raw read accession (SRA)	SRR39379362	SRR39379361	SRR39379360
Isolation source	California Botanic Garden soil	California Botanic Garden soil	California Botanic Garden soil
Coordinates	34°6'54"N 117°43'28"W	34°6'54"N 117°43'28"W	34°6'54"N 117°43'28"W
Elevation (m)	410	410	410
Adjacent plant	*Chilopsis linearis*	*Aesculus californica*	*Chilopsis linearis*
Sequencing platform	Oxford Nanopore (long-read)	Oxford Nanopore (long-read)	Oxford Nanopore (long-read)
Genome size (bp)	5,929,752	6,178,779	5,698,233
Contigs (n)	1	1	1
Topology	Circular	Circular	Circular
N50 (bp)	5,929,752	6,178,779	5,698,233
GC content (%)	62.6	62.4	62.6
Sequencing depth (×)	95	94	96
Total reads	559,553	354,501	546,337
Read N50 (bp)	8,803	9,050	9,663
Median read length (bp)	3,474	3,436	3,839
Median read quality (Phred)	24.52	25.79	23.48
Completeness % (CheckM)	99.74	99.74	100.00
Contamination % (CheckM)	0.87	1.94	0.20
Closest species (GTDB)	*Pseudomonas_E sp013179515*	Pseudomonas_E sp013179515	Pseudomonas_E capeferrum
ANI to closest (%)	99.75	99.53	98.66
Total genes	5,410	5,588	5,199
CDS (protein-coding genes)	5,264	5,449	5,055
tRNA	73	74	79
rRNA	22	22	22
ncRNA	50	42	42
Pseudogenes	10	11	12
Hypothetical proteins	177	233	172
Coding density (%)	89.7	89.6	90.9